

Update on Project 7: Security

Cody Gunton - August 12, 2026

<https://codygunton.github.io/talks-and-writing/2026-08-12-zkevm-breakout/>



RISC-V Compliance Testing

<https://github.com/eth-act/zkevm-test-monitor/>

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- Added tests of GPU provers for OpenVM and SP1
- OpenVM 2.1 is fully compliant with the standard target RV64IM_Zicclsm
- SP1 still has gaps
- Upgraded to testing Zisk 1.0.0-alpha; one gap that is patched in 1.0.0-beta 🙄

zkvmBlast

Stefanos Chaliasos, Martín Ochoa and Varun Thakore published a RISC-V ZKVM fuzzing library with the help of an EF grant

- Some contributions: seed corpus; Cascade-style RISC-V generation and random Rust program generation; fuzzer for several ZKVM proving pipelines (including GPU proving)
- 7 bugs found and reported (and counting)
- Blog post: <https://blog.zksecurity.xyz/posts/zkvmblast/>
- Repo: <https://github.com/zksecurity/zkvmblast>
- Seeds: <https://github.com/zksecurity/riscv-seeds-lab>

ZKVM Handbook

Ignacio and I wrote a first draft of our ZKVM and guest program assessment frameworks:

<https://github.com/eth-act/zkevm-standards/blob/main/handbooks/zkvm-handbook.md>

- Conform to eth-act standards; pass tests
- Reproducible sufficient performance on reference hardware; reasonable proof sizes
- Conform to cryptographic protocol and formal verification requirements
- Permissive licensing

Formally proving RISC-V Compliance of Zisk

<https://github.com/eth-act/zisk-fv>

- Found two more circuit bugs while tightening hypotheses ([#1217](#), [#1228](#))
- Still winding down solo work (50 PRs...) / ramping up to collaboration with Zisk
- Discovered some dropped constraints by round-tripping (would have discovered in a root theorem API audit)

Improvements in Verified-zkEVM libraries

- [CompPoly](#): shipping pre-built artifacts for faster development [301](#), faster BN254 and BLS12 arithmetic [284](#); more KoalaBear extensions [282](#), [283](#) and support for binary fields [278](#).
- [ArkLib](#): Progress on WHIR formalization [534](#), [599](#), [603](#), [616](#), [658](#), [667](#)